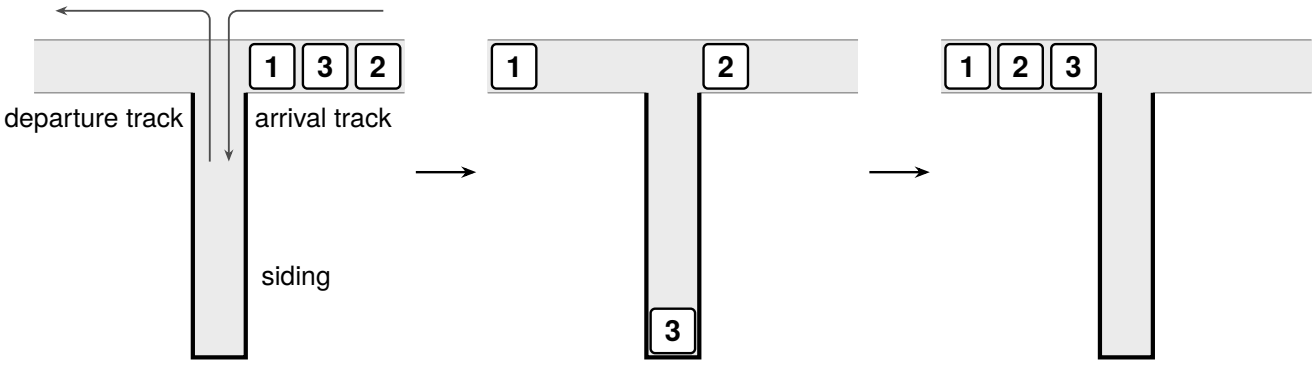
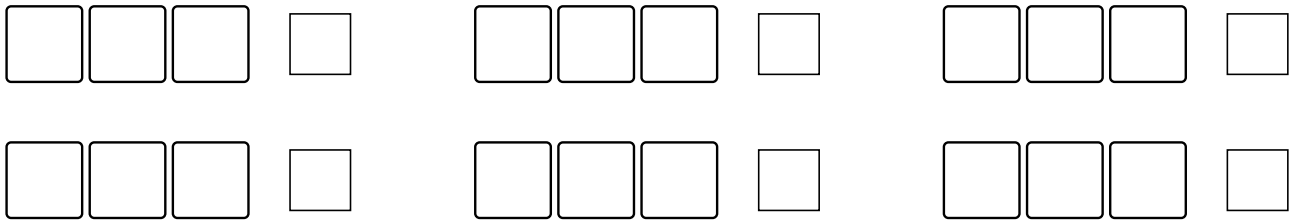
Put a train on the arrival track exactly as it is drawn, so the car on the left arrives first. Each move sends the front car of the arrival track down into the siding, or sends the top car of the siding out to the departure track. Cars slide along the table and never jump over each other. A car in the siding cannot go back to the arrival track. A train can *leave in order* if its cars can reach the departure track in the order 1, 2, 3, and so on.



Problem 1: Write every different train you can make with cars 1, 2, 3. Mark each one ✓ if it can leave in order and ✗ if it cannot.



Problem 2: Cars 1, 2, 3, 4 can be lined up as 24 different trains. With the other pair at your table, write each of these trains on a card, drive it, and sort the cards into a pile of trains that can leave in order and a pile of trains that cannot. How many cards are in each pile?

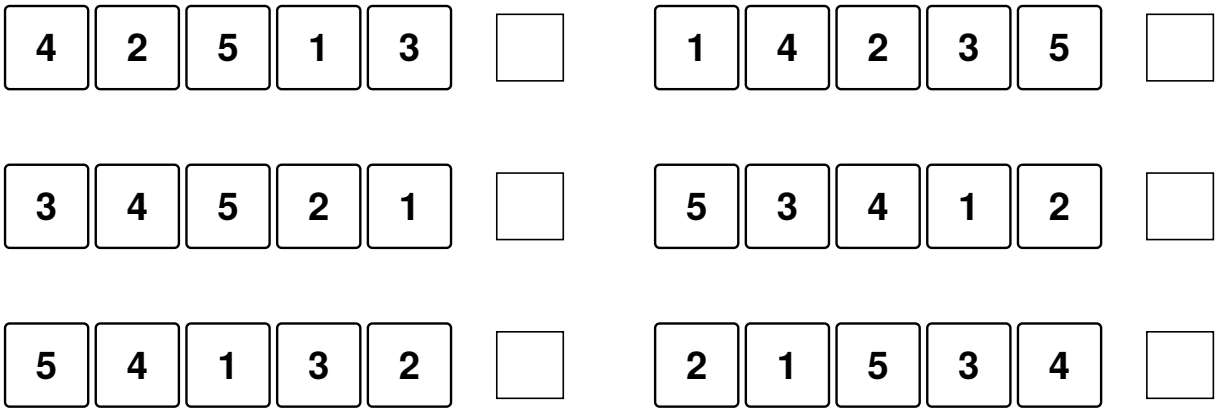
Can leave in order: _____ Cannot: _____

Problem 3: Which of these trains can leave in order? Mark each one ✓ or ✗.

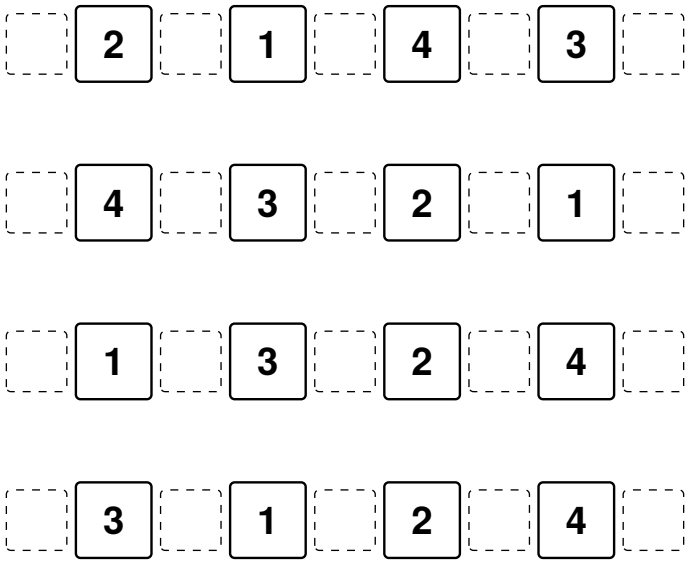
3	2	1	5	4	☐	4	1	3	2	5	☐
2	4	3	5	1	☐	1	2	5	3	4	☐
4	5	3	2	1	☐	3	1	5	4	2	☐
5	2	1	4	3	☐	1	5	3	4	2	☐

Problem 4: What makes a train impossible? Write a way to tell, without driving the cars, whether a train can leave in order.

Problem 5: Decide without driving which of these trains can leave in order, and mark each one ✓ or ✗. Then drive them to check.



Problem 6: Each of these trains can leave in order. Write 5 in every dashed box where car 5 could join the train so that it still leaves in order. How can you tell where car 5 can go?



Problem 7: How many of the 120 trains with cars 1, 2, 3, 4, 5 can leave in order? Explain how you know your count is right.